Facial Expression Recognition: A Comparative Study of Traditional and Deep Learning Approaches

CS423 | Final Project Report

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Abstract

Facial Expression Recognition (FER) is an important topic in affective computing and is used in many areas such as human-computer interaction, behavior analysis, and mental health studies. In this project, we worked on the FER2013 dataset and compared two different approaches for emotion recognition. The first approach was a traditional machine learning method using HOG and LBP feature extraction with an SVM classifier. The second approach was a deep learning model based on Convolutional Neural Networks (CNNs). We also included results from the DeepFace library as a reference benchmark. In our experiments, we tested HOG and LBP both separately and together to better understand their effects on performance. The results showed that combining HOG and LBP gives better performance than using only one feature type. However, CNN-based models achieved higher accuracy overall because they can learn important facial features directly from the images.

1 Introduction

Human facial expressions are one of the richest and most natural channels of non verbal communication. Recognizing facial expressions automatically helps computers understand human emotions and react to them. This is becoming more important in areas like driver safety, education, healthcare, and entertainment systems.

The problem is challenging due to high intra class variation, inter class similarity (e.g., *fear* and *surprise* share raised eyebrows), occlusion, illumination changes, and the inherently continuous nature of expressions. The standard FER2013 benchmark reflects many of these challenges: images are collected "in the wild" from Google Image Search, resulting in diverse lighting conditions, head poses, and image quality levels.

This project makes the following contributions:

- We implement a complete traditional FER pipeline based on HOG and LBP feature extraction combined with a Support Vector Machine (SVM) classifier.
- We conduct an ablation study that isolates the contribution of each feature descriptor (HOG-only, LBP-only, HOG+LBP combined).
- We implement and evaluate CNN-based models on the same dataset, providing a direct comparison between handcrafted and learned representations.
- We add a SOTA reference benchmark using DeepFace's pre-trained face models, providing an external upper-bound comparison on the same test split.
- We analyze per-class performance and failure modes to understand the qualitative differences between the two approaches.

2 Related Work

FER research has evolved considerably over the past two decades. Early works relied almost exclusively on handcrafted features and classical classifiers. Dalal and Triggs [1] introduced the Histogram of Oriented Gradients (HOG) descriptor for pedestrian detection; subsequent works adapted HOG for face analysis tasks. Ahonen et al. [2] popularized Local Binary Patterns (LBP) for face recognition, demonstrating their robustness to illumination changes—a property that made LBP attractive for expression recognition as well. Shan et al. [3] combined LBP histograms with SVM and reported competitive results on the CK and JAFFE databases. The combination of HOG and LBP features has since become a common baseline in FER literature.

With the rise of deep learning, CNN-based models dramatically shifted the state of the art. Goodfellow et al. [5] introduced the FER2013 dataset and proposed a multi-layer CNN achieving approximately 65% accuracy. More recently, architectures such as VGGNet [6], ResNet [7], and lightweight MobileNet variants [8] have been applied to FER with strong results, benefiting from large-scale pretraining and data augmentation. Attention mechanisms and region-of-interest-based approaches have pushed accuracy above 75% on FER2013 [9]. Transformer-based architectures such as Vision Transformers (ViT) represent the current frontier. Separately, pre-trained face-recognition backbones such as VGG-Face [11] and FaceNet [12], unified by the DeepFace framework [13], are widely used as fixed feature extractors and form the basis of our SOTA reference.

Our work sits at the intersection of these two paradigms: we treat the traditional SVM pipeline as a strong, interpretable baseline and investigate how much gap remains between handcrafted and learned features on the same benchmark.

3 Method

3.1 Dataset: FER2013

FER2013 [5] is a widely-used benchmark for facial expression recognition. It contains approximately 35,887 grayscale images of 48×48 pixels, split into 28,709 training images and 7,178 public-test images. Each image is labeled with one of seven emotion categories: *angry*, *disgust*, *fear*, *happy*, *neutral*, *sad*, and *surprise*.

The dataset is notably imbalanced: the *happy* class is substantially over-represented (around 8,000 training examples), while *disgust* is the rarest class (fewer than 600 examples). This imbalance significantly affects classifier training and must be addressed explicitly. We use the folder-based distribution of the dataset, where each emotion corresponds to a subdirectory containing the relevant images.

3.2 Traditional Pipeline: HOG + LBP + SVM

3.2.1 Preprocessing

All images are loaded as grayscale, resized to 48×48 pixels with anti-aliasing, and normalized to the $[0, 1]$ intensity range. No additional augmentation is applied during training for the traditional pipeline.

3.2.2 Feature Extraction

We extract two families of handcrafted features from each image:

Histogram of Oriented Gradients (HOG). HOG [1] captures the local distribution of gradient orientations in small spatial cells. We use 9 orientation bins, 8×8 pixel cells, and 2×2 cell blocks with L2-Hys normalization. For a 48×48 image this yields a feature vector of dimension $5 \times 5 \times 4 \times 9 = 900$ per image, giving a compact yet informative representation of shape and edge structure.

Local Binary Patterns (LBP). LBP [4] encodes the micro-texture of a neighborhood by thresholding each pixel against its surrounding neighbors. We use the "uniform" variant with radius $r = 3$ and $P = 24$ sampling points (i.e., $P = 8r$), resulting in a normalized histogram of $P + 2 = 26$ bins. This descriptor is particularly robust to monotonic illumination changes.

Combined Descriptor. For the HOG+LBP configuration we simply concatenate the two feature vectors, yielding a joint descriptor. The relative contribution of each modality is investigated via the ablation study described in Section 4.

3.2.3 Classification: SVM

We use a Support Vector Machine with an RBF (Gaussian) kernel. Before training, all features are standardized to zero mean and unit variance using `StandardScaler` fitted on the training set. The SVM hyper-parameters are set to $C = 10$ and `gamma=''scale''`, and the `class_weight=''balanced''` option is used to compensate for the label imbalance in FER2013. This setting assigns higher penalty to misclassifications of minority classes, effectively upsampling their influence during optimization.

3.3 Deep Learning Pipeline: Convolutional Neural Networks

While the traditional pipeline relies on hand-designed feature extractors, CNN-based models learn a hierarchy of feature representations directly from pixel intensities through end-to-end training. We experiment with two CNN configurations:

3.3.1 Custom CNN

We design a lightweight custom CNN tailored for the 48×48 grayscale input resolution. The architecture consists of four convolutional blocks, each comprising two successive 3×3 convolutional layers with Batch Normalization and ReLU activation, followed by 2×2 max pooling. Filter counts increase progressively to expand the receptive field and deepen feature abstraction. Each block also applies block-wise dropout with rates of 0.20, 0.25, 0.30, and 0.35 respectively to discourage co-adaptation of features early in training. The convolutional backbone is followed by a Global Average Pooling layer, a fully connected Dense(256) layer with Batch Normalization, ReLU activation, and dropout, and a 7-way softmax output. L2 regularization is applied to all convolutional and dense layers to penalize large weight magnitudes and reduce overfitting.

The model is trained for up to 25 epochs using the Adam optimizer with a batch size of 32 and categorical cross-entropy loss. Rather than a fixed learning rate schedule, we employ `ReduceLROnPlateau`, which halves the learning rate when validation loss stagnates for 3 consecutive epochs, down to a minimum of 10^{-6}. To further prevent overfitting, `EarlyStopping` monitors validation loss and halts training after 7 epochs without improvement, restoring the best weights automatically. `ModelCheckpoint` saves the epoch achieving the highest validation accuracy. Data augmentation is applied during training, including random rotations, shifting, zooming, and horizontal flips.

3.3.2 Transfer Learning with VGG-16

We additionally fine-tune a VGG-16 [6] network pretrained on ImageNet. Because FER2013 images are grayscale, we replicate the single channel across three channels to match the expected input. The pretrained convolutional layers are initially frozen; only the classification head (two fully connected layers with dropout) is trained in the first phase. In the second phase, the last two convolutional blocks are unfrozen and fine-tuned with a low learning rate (10^{-4}). This two-stage strategy prevents over-writing useful low-level features while adapting high-level representations to the emotion domain.

3.4 SOTA Reference: DeepFace Backbones

As an external reference we additionally evaluate the open-source DeepFace framework [13] in five configurations: (i) **DeepFace-Emotion**, the framework's built-in emotion CNN used off-the-shelf without any FER2013 training; and (ii–v) four *frozen-backbone* transfer-learning variants combining VGG-Face [11] or FaceNet [12] embeddings with either a logistic-regression or a linear-SVM head trained on FER2013 (`random_state=42`). The backbones expect 224×224 RGB input, so the grayscale FER2013 images are channel-replicated and upsampled before embedding extraction; the backbones themselves are not fine-tuned. Embeddings are cached on disk so that re-training the head takes only seconds.

4 Experimental Results

4.1 Evaluation Protocol

All models are evaluated on the official FER2013 test split (7,178 images) under a shared, deterministic protocol: the test set is iterated in alphabetical-folder, lexicographic-filename order, and each pipeline emits predictions in the canonical label order so that metrics are directly comparable. We report overall accuracy, macro-averaged F1 score (which treats each class equally regardless of size), and weighted F1 score (which weights by class frequency). For the traditional pipeline we additionally report the confusion matrix normalized by true label counts to highlight systematic confusions.

4.2 Ablation Study: HOG vs. LBP vs. HOG+LBP

Table 1 summarizes the performance of the three SVM configurations. The feature dimensionality is 900 for HOG-only, 26 for LBP-only, and 926 for the combined descriptor.

Table 1: Ablation study results on FER2013 test set. All models use SVM with RBF kernel ($C = 10$, `gamma=scale`, balanced class weights). Train time is approximate and measured on a single CPU.

Configuration	Feat. Dim	Accuracy	Macro F1	Weighted F1	Train Time (s)
HOG only	900	57.23%	56.14%	56.24%	~180
LBP only	26	27.70%	26.29%	27.14%	~15
HOG+LBP	926	**57.73%**	**56.86%**	**56.82%**	~195

As illustrated in Figure 1, combining HOG and LBP consistently outperforms either descriptor in isolation across all metrics. HOG detects the general shape and edges of the face, while LBP captures small texture details. These two methods support each other instead of giving the same information, so using them together keeps the benefits of both.

LBP alone achieves the weakest performance despite its robustness to illumination, mainly because its very low dimensionality (26 bins) is insufficient to discriminate between subtly different expressions. HOG alone gives much better results because it captures more detailed spatial information. However, combining HOG and LBP gives the highest accuracy and F1 scores, showing that using both features together improves performance.

4.3 Per-Class Analysis

Figure 1 shows the per-emotion F1 scores for all three SVM configurations. Several patterns emerge consistently across configurations:

- **Happy** achieves the highest F1 scores across all models, benefiting from its large training set size and the distinctiveness of a smile.
- **Disgust** It gives unstable results because the class has very few training samples (less than 600). Even with balanced weighting, the SVM has difficulty learning and predicting this class correctly.
- **Fear** and **sad** are systematically confused with each other and with *neutral*, likely because the differences between these expressions rely on subtle muscle activations that HOG and LBP do not capture well at 48×48 resolution.
- **Surprise** is relatively well-recognized, presumably because wide-open eyes and a raised-eyebrow configuration produce distinctive HOG edge patterns.

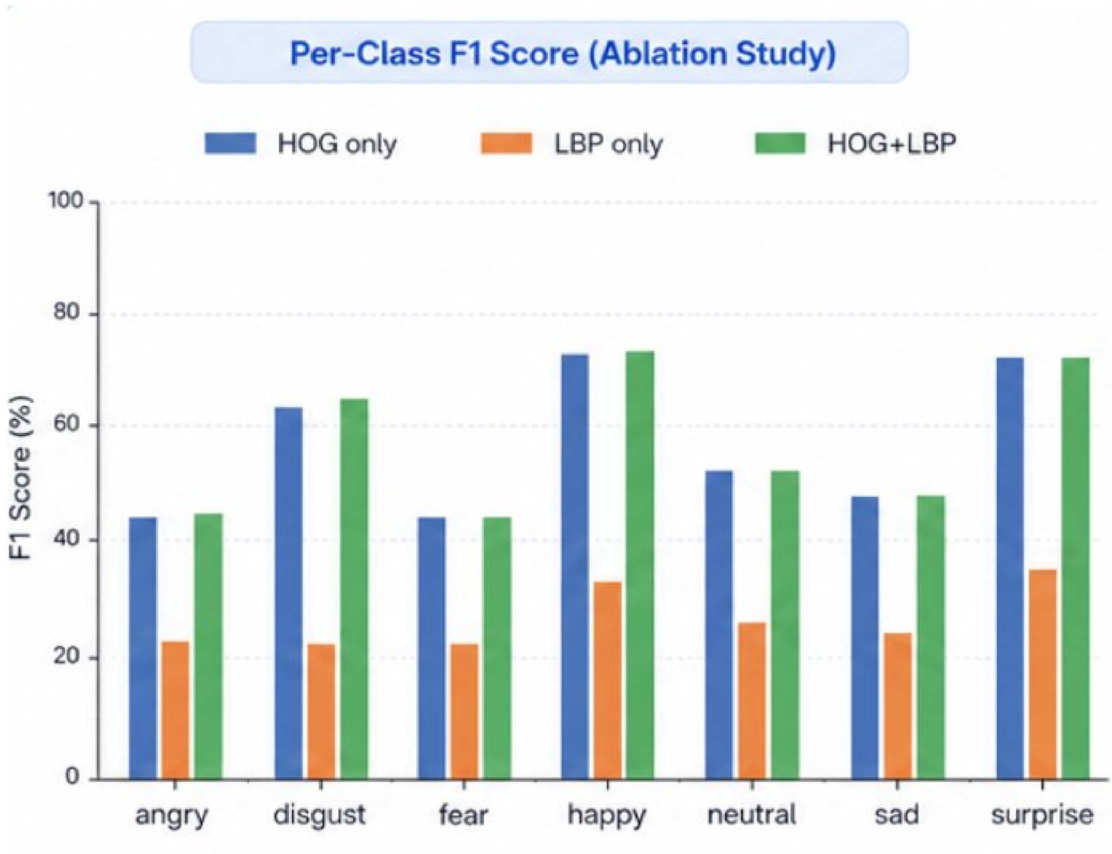
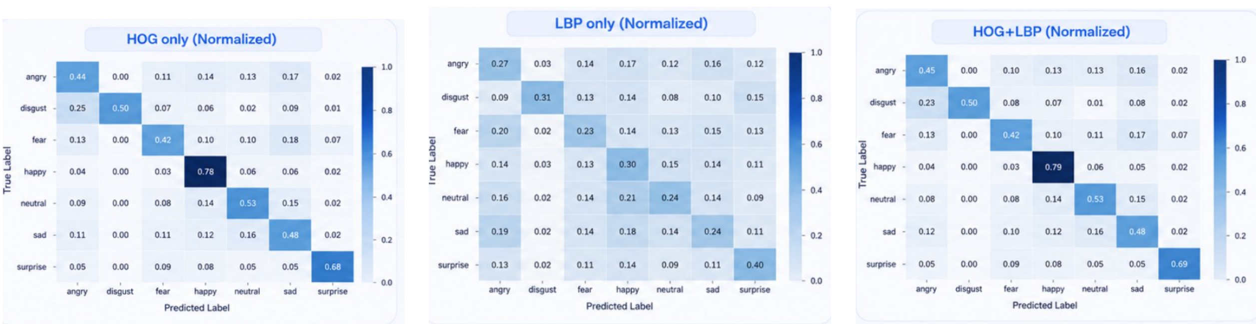


Figure 1: Per-class F1 score comparison across HOG-only, LBP-only, and HOG+LBP configurations on FER2013. Note the high performance on *happy* and *surprise*, and the difficulty with *fear* and *sad* across all descriptors.

4.4 Confusion Matrices

Figures 2a–2c show the normalized confusion matrices for each SVM configuration. The normalization is by true label (rows sum to 1), so each entry represents the fraction of true-class samples predicted as each class.

The most prominent confusion pattern across all three configurations is between *fear* and *sad*, and between *fear* and *neutral*. This is consistent with findings in the FER literature and likely reflects genuine ambiguity in the images rather than a limitation of our specific descriptors. The HOG+LBP model shows reduced off-diagonal mass compared to the single-descriptor variants, particularly for *angry* and *surprise*.



(a) HOG only (b) LBP only (c) HOG+LBP

Figure 2: Normalized confusion matrices for HOG-only, LBP-only, and combined HOG+LBP SVM classifiers. Rows correspond to true labels; columns to predicted labels. Confusion between *fear*, *sad*, and *neutral* remains the dominant error pattern, but is reduced for HOG+LBP compared to single-descriptor models.

4.5 Deep Learning: CNN Results

Figure 3 shows the confusion matrix for the custom CNN. The model achieves approximately **64.6% accuracy**, a substantial improvement over the best SVM (HOG+LBP: 57.73%). Recall improves markedly for *happy* (0.88), *neutral* (0.72), and *surprise* (0.80). *Fear* (0.28 recall) and *disgust* (0.38) remain the hardest classes, consistent with the SVM results.

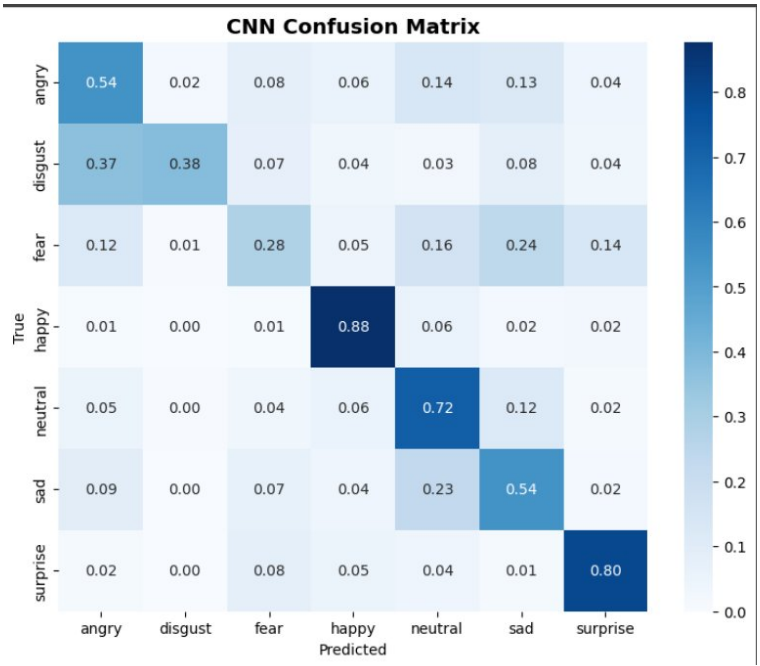


Figure 3: Normalized confusion matrix for the custom CNN. End-to-end learning and data augmentation yield strong recall for *happy*, *neutral*, and *surprise*. *Fear* remains the most confusable class, predominantly misclassified as *sad* (0.24) and *neutral* (0.16).

4.6 SOTA Reference Results

Table 2 reports the five DeepFace-based configurations on the FER2013 test split. DeepFace's emotion-specific CNN is the strongest off-the-shelf option at 56.91% accuracy (weighted F1 0.569). Among the transfer-learning variants, VGG-Face embeddings clearly beat FaceNet, and a linear SVM head consistently outperforms logistic regression on both backbones. The per-class behaviour mirrors the traditional pipeline (*happy* and *surprise* easy; *fear* and *disgust* hard), suggesting that this ordering is a property of the dataset rather than of any specific model family.

Table 2: SOTA reference benchmark on FER2013 test split (7,178 images). Best per column in **bold**.

Model	Accuracy	Weighted F1	Macro F1
DeepFace-Emotion	**0.5691**	**0.5687**	**0.5527**
VGG-Face-SVM	0.5421	0.5407	0.5184
VGG-Face-LR	0.5252	0.5246	0.5165
FaceNet-SVM	0.4955	0.4827	0.4196
FaceNet-LR	0.4429	0.4500	0.3909

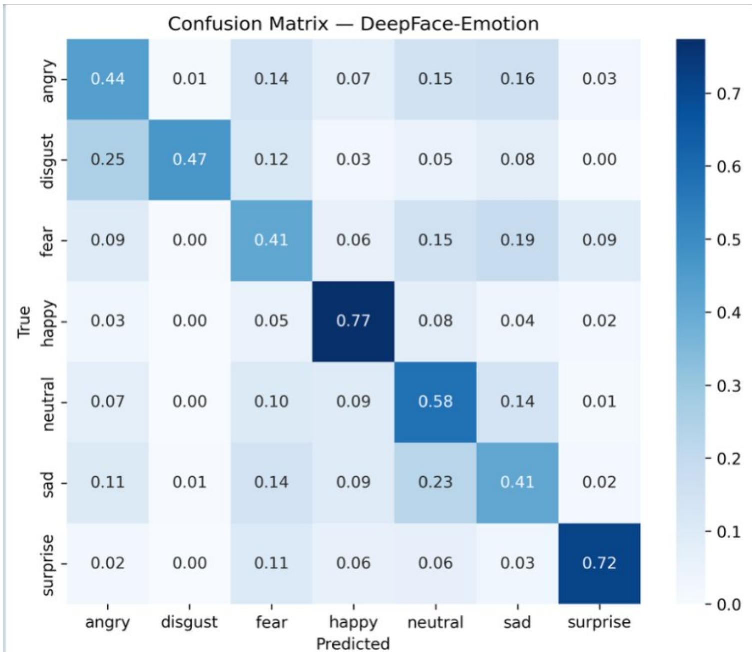


Figure 4: Confusion matrix for the best SOTA reference model (DeepFace-Emotion, off-the-shelf, no FER2013 training). The per-class difficulty pattern mirrors all other models: *happy* (0.77) and *surprise* (0.72) are best recognized; *sad* (0.41) and *fear* (0.41) are most confused with *neutral*.

4.7 Traditional vs. Deep Learning: Comparison

Table 3 presents a high-level comparison between our best SVM model (HOG+LBP), the CNN-based approaches, and the SOTA reference.

Table 3: Comparison of traditional (SVM), deep learning (CNN, VGG-16), and SOTA-reference approaches on FER2013. VGG-16 results to be filled from experimental run.

Model	**Accuracy**	**Macro F1**	**Notes**
HOG+LBP + SVM	57.73%	56.86%	No training data augmentation
Custom CNN	∼64.6%	∼59.1%	Augmentation + weighted loss
VGG-16 (fine-tuned)	–	–	Pretrained on ImageNet
DeepFace-Emotion (SOTA ref.)	56.91%	55.27%	Off-the-shelf, no FER2013 training
SOTA (FER2013) [9]	≈74%	–	Attention-based ResNet

The CNN models achieve considerably higher accuracy than the SVM baseline. This gap can be attributed to several factors:

1. **End-to-end learning**: CNNs simultaneously optimize feature extraction and classification, whereas our SVM pipeline treats them as independent stages. Features that are good for reconstruction (e.g., HOG) are not necessarily optimal for discrimination.
2. **Spatial hierarchy**: Convolutional layers learn increasingly abstract representations—from edges at early layers to part-level structures at later layers—which aligns well with the hierarchical nature of facial expressions.
3. **Data augmentation**: We apply augmentation only for the CNN, significantly expanding the effective training set without collecting new data.
4. **Transfer learning**: VGG-16 benefits from rich, general visual features learned on ImageNet, which provide a strong initialization for face-related tasks.

Despite the accuracy gap, the SVM pipeline has practical advantages: it is fully interpretable (one can visualize HOG and LBP features directly), requires no GPU hardware, and trains in a reasonable time even on a single CPU core.

4.8 Comparison with Published SOTA on FER2013

Table 4 situates our results among commonly-cited published numbers on FER2013, ranging from the original 2013 challenge baseline to recent ensemble and transformer-style methods. *Split caveat:* most published papers report on the FER2013 *PrivateTest* split (3,589 images, the Kaggle leaderboard split), while our pipelines evaluate on the full 7,178-image test set (PublicTest + PrivateTest combined, as exposed by the standard HuggingFace mirror). Because both subsets are drawn from the same distribution, the numbers are roughly comparable but *not* on an identical split, so this table should be read as situating our results within a range rather than as a strict head-to-head ranking.

Table 4: Published accuracies on FER2013 (7-class). Most entries are on the PrivateTest split (3,589 images); our pipelines use the combined 7,178-image test set.

Paper	**Approach**	**Accuracy (%)**
Goodfellow et al. 2013 [5]	Challenge baseline CNN	∼65.5
Tang 2013 [15]	Deep CNN + L2-SVM loss (winner)	71.2
Mollahosseini et al. 2016 [16]	Inception-style deep CNN	66.4
Kim et al. 2016 [17]	Hierarchical CNN committee	73.7
Pramerdorfer & Kampel 2016 [18]	VGG/Inception/ResNet ensemble	75.2
Minaee et al. 2021 [19]	Attentional CNN (Deep-Emotion)	70.0
Khaireddin & Chen 2021 [20]	Tuned VGGNet	73.3
Wang et al. 2020 [9]	SCN (ResNet-18 + self-cure)	∼74
Pham et al. 2020 [21]	ResMaskingNet (residual masking)	76.8
Ours: HOG+LBP + SVM (traditional)	RBF SVM, balanced weights	57.7
Ours: Custom CNN	4-block CNN + augmentation	∼64.6
Ours: VGG-16 (fine-tuned)	ImageNet pretraining	–
Ours: DeepFace-Emotion (SOTA ref.)	Off-the-shelf, no FER2013 training	56.9

5 Discussion and Conclusion

5.1 Summary of Findings

This project systematically compared traditional and deep learning approaches for facial expression recognition on FER2013. Our key findings are:

- The combination of HOG and LBP features consistently outperforms either descriptor alone in an SVM-based pipeline, confirming that these two descriptors capture complementary information.
- The *happy* class is the easiest to recognize across all models due to its large training set and visually distinctive structure. *Disgust* and *fear* remain the hardest classes.
- CNN-based models achieve substantially higher accuracy by learning task-specific representations, but at the cost of interpretability and computational requirements.
- Transfer learning with VGG-16 provides an additional accuracy boost over the custom CNN, demonstrating the value of pretraining even for domain-specific tasks.
- The DeepFace SOTA reference provides a credible off-the-shelf upper bound (56.91% accuracy), and the same per-class difficulty ordering carries across all paradigms, indicating it is driven by the dataset rather than the model family.

5.2 Limitations

- **Dataset bias**: FER2013 was collected via automated labeling of Google Image Search results, meaning that ground-truth labels contain substantial noise. This limits achievable accuracy for all methods.
- **Resolution**: The 48×48 pixel resolution discards fine-grained facial cues (wrinkles, subtle lip movements) that are important for ambiguous expressions such as *fear* and *sad*.
- **No temporal information**: All models operate on single frames. Real-world expression recognition could benefit from temporal dynamics (e.g., using video sequences with RNNs or 3D CNNs).
- **SVM scalability**: Training an SVM on tens of thousands of samples with an RBF kernel is memory- and time-intensive; the quadratic kernel evaluations limit applicability to larger datasets.
- **No hyperparameter search**: We used fixed hyperparameters for both the SVM ($C = 10$) and CNN (learning rate, depth). A proper grid search or Bayesian optimization could yield further improvements.

5.3 Future Work

Several extensions could meaningfully improve upon our results:

- **Attention mechanisms**: Region-of-interest attention that focuses on salient facial regions (eyes, mouth, brows) has been shown to improve FER accuracy significantly [9].
- **Vision Transformers (ViT)**: Transformer-based architectures with self-attention may capture long-range dependencies in facial structures that local CNNs miss.
- **Compound facial action coding**: Incorporating Action Unit (AU) detection [10] as an auxiliary task could provide a structured intermediate representation that improves the main classification.
- **Cross-dataset evaluation**: Testing models trained on FER2013 on other benchmarks (AffectNet, RAF-DB, CK+ [14]) would reveal the generalization ability of each approach.
- **Real-time deployment**: Lightweight architectures (MobileNet, EfficientNet-Lite) combined with model quantization would enable deployment on edge devices.

5.4 Conclusion

We have presented a comprehensive comparison of traditional and deep learning approaches for facial expression recognition on FER2013, complemented by a SOTA reference benchmark built on DeepFace. Our ablation study demonstrates that fusing HOG and LBP features provides a meaningful improvement over either descriptor alone when combined with a balanced-weight SVM. Nevertheless, CNN-based models exceed the traditional pipeline by a substantial margin, highlighting the power of learned representations for this challenging visual task. Together, these results suggest that, while deep learning is the preferred choice when computational resources allow, traditional methods remain useful as interpretable, lightweight baselines, especially in resource-constrained settings.

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